

JIWON KIM

PhD Candidate, Neuroscience · University of Pittsburgh | Pittsburgh, PA | jik123@pitt.edu

EDUCATION

University of Pittsburgh

Jun 2023 – Present

PhD Candidate, Neuroscience · Center for Neuroscience (CNUP), Pittsburgh, PA

- Dissertation: Cortical interneuron regulation of angiogenesis and neurovascular coupling in neurodegeneration
- Advisors: Dr. Alberto Vazquez & Dr. Sarah Ross | Passed Comprehensive Exam, Dec 2025
- Teaching Assistant, Brain and Behavior (150 students), Fall 2024

Washington and Lee University

Sep 2016 – May 2020

BS, Neuroscience · Lexington, VA | GPA: 3.79

- Omicron Delta Kappa National Leadership Honor Society; Phi Xi Tri-Beta National Biological Honor Society; Phi Eta Sigma First-Year Honor Society; President's List

RELEVANT EXPERIENCE & ACCOMPLISHMENTS

PhD Candidate — Vazquez & Ross Labs, University of Pittsburgh

Jun 2023 – Present

- Designed and executed an independent research project from hypothesis to preliminary data, generating results that secured \$70,676 in AHA Predoctoral Fellowship funding (3.66th percentile)
- Investigating cortical interneuron regulation of angiogenesis and neurovascular coupling using transgenic mouse models, 2P in vivo imaging, and molecular techniques — with direct implications for stroke recovery and neurodegeneration therapeutics
- Contributed key experiments to a cross-lab collaboration resulting in a co-authored publication in *Science* (2025), communicating findings across teams with distinct scientific focuses
- Presented thesis work and rotation findings at 3 conferences over 2 years; awarded \$1,400 competitive travel bursary

Research Assistant — Yuan Lab, Boston Children's Hospital / Harvard Medical School

Aug 2020 – May 2022

- Contributed to 4 published papers across 3 simultaneous projects within a 5-person team, collaborating closely with external labs and clinicians to align experimental findings with patient-relevant outcomes in pulmonary vascular disease
- Drove an independent project on pericyte-mediated vascular remodeling in CTD-PAH from bench to manuscript, authoring the original draft that contributed to a publication in *EMBO Reports* (2024)
- Co-developed and optimized a precision-cut lung slice methodology adopted as a published, reproducible protocol (*J. Vis. Exp.*, 2021), translating a technical workflow into a shareable tool for the broader research community
- As one of the first members of a newly established lab, built operational infrastructure while conducting independent research — managing competing priorities across experimental and administrative responsibilities

GRANTS & AWARDS

- American Heart Association Predoctoral Fellowship — \$70,676 (3.66th percentile), Jan 2026 – Dec 2027
- Center for International Studies Summer Grant (\$5,000) — Washington and Lee University, 2019
- Goldman Sachs Fund (\$5,514) — Washington and Lee University, 2018
- Endeavor Foundation Grant (\$4,250) — Washington and Lee University, 2017

LEADERSHIP

President — Young Adult Group, Korean Central Church of Pittsburgh

- Led a 100+ member young adult ministry comprising 4 teams and 8 small groups
- Managed an annual budget of \$15,000 across programming, outreach, and operations *Apr 2025 – Apr 2026*
- Directed full-cycle event planning and execution for large-scale community and worship events
- Coordinated cross-functional leadership teams to align goals and drive community engagement

Founder & Director — Vision Unification (Nonprofit)

- Secured \$5,000 in grant funding to launch a nonprofit promoting cultural unification among Koreans
- Scaled the organization's mission to support North Korean defectors and students interested in Korean reunification, sustaining operations for 6+ years *May 2019 – Present*

TECHNICAL SKILLS

In Vivo Models	Photothrombotic stroke, craniotomy & cranial window preparation, stereotaxic injections, retroorbital injections, hypoxia chamber maintenance & monitoring, intraperitoneal injections, transgenic mouse husbandry
Imaging	Two-photon (2P) in vivo imaging, widefield imaging with whisker stimulation, laser speckle contrast imaging, immunofluorescence/immunostaining, cryosectioning, tissue clearing, in situ fluorescence (ISH), MRI/neuroimaging analysis
Molecular Biology	Single-cell RNA sequencing (scRNA-seq), RNA isolation, FACS/cell sorting, tissue harvesting & sectioning, precision-cut lung slices (PCLS), in vitro cell experiments
Data Analysis	MATLAB, Python, R, SPSS, SAS, E-Prime, systematic literature review (PRISMA)
Scientific Communication	Peer-reviewed manuscripts (first & contributing author), grant writing, poster & oral presentations, cross-disciplinary team collaboration

PUBLICATIONS

- A.W. Liu, Y.R. Zhang, ..., **J. Kim**, T.D. Sheahan, S.E. Ross, M. Meisel, T.L. Sumpter, and D.H. Kaplan. "Scratching promotes allergic inflammation and host defense via neurogenic mast cell activation." *Science*, 387(6733):eadn9390, 2025.
- H. Kim, Y. Liu, **J. Kim**, et al. "Pericytes contribute to pulmonary vascular remodeling via HIF2 α signaling." *EMBO Reports*, 2024.
- E. Taglauer, E. Wachman, ..., **J. Kim**, ..., K. Yuan. "Acute SARS-CoV-2 Infection in Pregnancy Is Associated with Placental ACE2 Shedding." *The American Journal of Pathology*, 2022.
- T. Klouda*, Y. Hao*, H. Kim, **J. Kim**, et al. "Interferon-alpha Facilitates SARS-CoV-2 Pulmonary Vascular Infection by Inducing ACE2." *Angiogenesis*, 2021.
- T. Klouda, H. Kim, **J. Kim**, G. Visner, K. Yuan. "Precision Cut Lung Slices as an Efficient Tool for Ex Vivo Pulmonary Vessel Studies." *Journal of Visualized Experiments*, 2021.